RAFFLES INSTITUTION
H2 Mathematics (9758)
2025 Year 5

Tutorial 9A: Integration Techniques

Section A (Basic Questions)

Evaluate the following integrals

1. (a) $\int (5x-2)^6 \, \mathrm{d}x$ (b) $\int \frac{4}{3-2x} \, \mathrm{d}x$

(c) $\int \mathrm{e}^{1-2x} \, \mathrm{d}x$ (d) $\int \cos(3x+1) \, \mathrm{d}x$

a) $\int (5x-2)^6 \, \mathrm{d}x = \frac{1}{5}\int 5(5x-2)^6 \, \mathrm{d}x = \frac{1}{5}\frac{(5x-2)^7}{7} + c = \frac{1}{35}(5x-2)^7 + c$

b) $\int \frac{4}{3-2x} \, \mathrm{d}x = \frac{4}{-2}\int \frac{-2}{3-2x} \, \mathrm{d}x = -2\ln|3-2x| + c$

c) $\int \mathrm{e}^{1-2x} \, \mathrm{d}x = -\frac{1}{2}\int (-2)\mathrm{e}^{1-2x} \, \mathrm{d}x = -\frac{1}{2}\mathrm{e}^{1-2x} + c$

d) $\int \cos(3x+1) \, \mathrm{d}x = \frac{1}{3}\int 3\cos(3x+1) \, \mathrm{d}x = \frac{1}{3}\sin(3x+1) + c$

2. (a) $\int \frac{1}{(1-4x)^{\frac{5}{3}}} \, \mathrm{d}x$ (b) $\int \frac{x}{2x^2+3} \, \mathrm{d}x$

(c) $\int (\mathrm{e}^x - 3\mathrm{e}^{2x})^2 \, \mathrm{d}x$ (d) $\int \sec^2 \frac{x}{3} \, \mathrm{d}x$

a) $\int \frac{1}{(1-4x)^{\frac{5}{3}}} \, \mathrm{d}x = -\frac{1}{4}\int (-4)(1-4x)^{-\frac{5}{3}} \, \mathrm{d}x = -\frac{1}{4}\frac{(1-4x)^{-\frac{2}{3}}}{-\frac{2}{3}} + c = \frac{3}{8}(1-4x)^{-\frac{2}{3}} + c$

b) $\int \frac{x}{2x^2+3} \, \mathrm{d}x = \frac{1}{4}\int \frac{4x}{2x^2+3} \, \mathrm{d}x = \frac{1}{4}\ln\left(2x^2+3\right) + c$

c) $\int (\mathrm{e}^x - 3\mathrm{e}^{2x})^2 \, \mathrm{d}x = \int (\mathrm{e}^{2x} - 6\mathrm{e}^{3x} + 9\mathrm{e}^{4x}) \, \mathrm{d}x = \frac{1}{2}\mathrm{e}^{2x} - 2\mathrm{e}^{3x} + \frac{9}{4}\mathrm{e}^{4x} + c$

d) $\int \sec^2 \frac{x}{3} \, \mathrm{d}x = 3\int \frac{1}{3}\sec^2 \frac{x}{3} \, \mathrm{d}x = 3\tan\frac{x}{3} + c$

3. (a) $\int \frac{x^2-1}{\sqrt{x^3-3x}} \, \mathrm{d}x$ (b) $\int \left(x^2-1\right)\left(x^3-3x\right)^5 \, \mathrm{d}x$ (c) $\int \frac{x}{x+3} \mathrm{d}x$

(d) $\int \frac{1}{\sqrt{x}(1-\sqrt{x})} \, \mathrm{d}x$ (e) $\int \sin^2 3x \, \mathrm{d}x$ (f) $\int \tan 3x \, \mathrm{d}x$

(g) $\int \cot\theta \, \mathrm{d}\theta$ (h) $\int \sin\frac{x}{2}\cos\frac{x}{2} \, \mathrm{d}x$ (i) $\int \cos(3x)\sin^3(3x) \, \mathrm{d}x$

a) $\displaystyle\int \frac{x^2-1}{\sqrt{x^3-3x}}\ \mathrm{d}x = \frac{1}{3}\int \frac{3(x^2-1)}{\sqrt{x^3-3x}}\ \mathrm{d}x = \frac{1}{3}\frac{\left(x^3-3x\right)^{\frac{1}{2}}}{\frac{1}{2}} + c = \frac{2}{3}\left(x^3-3x\right)^{\frac{1}{2}} + c$
b) $\displaystyle\int \left(x^2-1\right)\left(x^3-3x\right)^5\ \mathrm{d}x = \frac{1}{3}\int 3\left(x^2-1\right)\left(x^3-3x\right)^5\ \mathrm{d}x = \frac{1}{18}\left(x^3-3x\right)^6 + c$
c) $\displaystyle\int \frac{x}{x+3}\mathrm{d}x = \int \frac{x+3-3}{x+3}\mathrm{d}x = \int \left(1-\frac{3}{x+3}\right)\mathrm{d}x = x - 3\ln\lvert x+3\rvert + c$
d) $\displaystyle\int \frac{1}{\sqrt{x}(1-\sqrt{x})}\ \mathrm{d}x = -2\int \frac{1}{-2\sqrt{x}}\frac{1}{(1-\sqrt{x})}\ \mathrm{d}x = -2\ln\left\lvert 1-\sqrt{x}\right\rvert + c$
e) $\displaystyle\int \sin^2 3x\,\mathrm{d}x = \frac{1}{2}\int \left(1-\cos 6x\right)\mathrm{d}x = \frac{1}{2}\left(x - \frac{\sin 6x}{6}\right) + c$
f) $\displaystyle\int \tan 3x\ \mathrm{d}x = -\frac{1}{3}\int \frac{-3\sin 3x}{\cos 3x}\ \mathrm{d}x = -\frac{1}{3}\ln\lvert\cos 3x\rvert + c$
g) $\displaystyle\int \cot\theta\,\mathrm{d}\theta = \int \frac{\cos\theta}{\sin\theta}\mathrm{d}\theta = \ln\lvert\sin\theta\rvert + c$
h) $\displaystyle\int \sin\frac{x}{2}\cos\frac{x}{2}\ \mathrm{d}x = \frac{1}{2}\int \sin x\ \mathrm{d}x = -\frac{1}{2}\cos x + c$
i) $\displaystyle\int \cos(3x)\sin^3(3x)\,\mathrm{d}x = \frac{1}{3}\int 3\cos(3x)\sin^3(3x)\,\mathrm{d}x = \frac{1}{3}\frac{\sin^4(3x)}{4} + c = \frac{1}{12}\sin^4(3x) + c$

4. (a) $\displaystyle\int \frac{\sec^2 x}{(1+\tan x)^3}\ \mathrm{d}x$ (b) $\displaystyle\int \frac{1}{2x^2-12x-14}\ \mathrm{d}x$ (c) $\displaystyle\int \frac{1}{x^2+2x}\ \mathrm{d}x$

(d) $\displaystyle\int \frac{x^3+2}{x^2-1}\ \mathrm{d}x$ (e) $\displaystyle\int \frac{2x^2-x+9}{(x+1)(x-3)^2}\ \mathrm{d}x$

a) $\displaystyle\int \frac{\sec^2 x}{(1+\tan x)^3}\ \mathrm{d}x = \int \sec^2 x\left(1+\tan x\right)^{-3}\ \mathrm{d}x = \frac{(1+\tan x)^{-2}}{-2} + c = -\frac{1}{2\left(1+\tan x\right)^2} + c$
b) $\displaystyle\frac{1}{(x-7)(x+1)} = \frac{A}{x-7} + \frac{B}{x+1}$ Then $1 = A(x+1) + B(x-7)$ Let $x=-1$, $\quad 1 = -8B \Rightarrow B = -\dfrac{1}{8}$ Let $x=7$, $\quad 1 = 8A \Rightarrow A = \dfrac{1}{8}$ $\displaystyle\therefore \frac{1}{(x-7)(x+1)} = \frac{1}{8}\left(\frac{1}{x-7} - \frac{1}{x+1}\right)$

Method 1:

$$\int \frac{1}{2x^2-12x-14}\ \mathrm{d}x$$
$$=\frac{1}{2}\int \frac{1}{x^2-6x-7}\ \mathrm{d}x$$
$$=\frac{1}{2}\int \frac{1}{(x-7)(x+1)}\ \mathrm{d}x$$
$$=\frac{1}{16}\int \left(\frac{1}{x-7}-\frac{1}{x+1}\right)\ \mathrm{d}x$$
$$=\frac{1}{16}\left(\ln|x-7|-\ln|x+1|\right)+c$$

Method 2:

$$\int \frac{1}{2x^2-12x-14}\ \mathrm{d}x$$
$$=\frac{1}{2}\int \frac{1}{(x-3)^2-4^2}\ \mathrm{d}x$$
$$=\frac{1}{2}\left(\frac{1}{2(4)}\right)\ln\left|\frac{(x-3)-4}{(x-3)+4}\right|+c$$
$$=\frac{1}{16}\ln\left|\frac{x-7}{x+1}\right|+c$$

c)

$$\frac{1}{x(x+2)}=\frac{A}{x}+\frac{B}{x+2}$$

Then $1=A(x+2)+Bx$

Let $x=-2$, $1=-2B \Rightarrow B=-\frac{1}{2}$

Let $x=0$, $1=2A \Rightarrow A=\frac{1}{2}$

$$\therefore \frac{1}{x(x+2)}=\frac{1}{2}\left(\frac{1}{x}-\frac{1}{x+2}\right)$$

Method 1:

$$\int \frac{1}{x^2+2x}\ \mathrm{d}x$$
$$=\int \frac{1}{x(x+2)}\ \mathrm{d}x$$
$$=\frac{1}{2}\int \left(\frac{1}{x}-\frac{1}{x+2}\right)\ \mathrm{d}x$$
$$=\frac{1}{2}\left(\ln|x|-\ln|x+2|\right)+c$$

Method 2:

$$\int \frac{1}{x^2+2x}\ \mathrm{d}x=\int \frac{1}{(x+1)^2-1}\ \mathrm{d}x$$
$$=\frac{1}{2}\ln\left|\frac{(x+1)-1}{(x+1)+1}\right|+c$$
$$=\frac{1}{2}\ln\left|\frac{x}{x+2}\right|+c$$

d) $\int \frac{x^3+2}{x^2-1}\ \mathrm{d}x$ $= \int \frac{x(x^2-1)+x+2}{x^2-1}\ \mathrm{d}x$ $= \int x + \frac{x+2}{(x+1)(x-1)}\ \mathrm{d}x$ $= \int x + \frac{-\frac{1}{2}}{x+1} + \frac{\frac{3}{2}}{x-1}\ \mathrm{d}x$ $= \frac{x^2}{2} - \frac{1}{2}\ln\lvert x+1\rvert + \frac{3}{2}\ln\lvert x-1\rvert + c$	$\frac{x+2}{(x+1)(x-1)} = \frac{A}{x+1} + \frac{B}{x-1}$ Then $x+2 = A(x-1)+B(x+1)$ Let $x=1$, $3 = 2B \Rightarrow B = \frac{3}{2}$ Let $x=-1$, $1 = -2A \Rightarrow A = -\frac{1}{2}$ $\therefore \frac{x+2}{(x+1)(x-1)} = \frac{-\frac{1}{2}}{x+1} + \frac{\frac{3}{2}}{x-1}$
e) $\int \frac{2x^2-x+9}{(x+1)(x-3)^2}\ \mathrm{d}x$ $= \int \frac{\frac{3}{4}}{x+1} + \frac{\frac{5}{4}}{x-3} + \frac{6}{(x-3)^2}\ \mathrm{d}x$ $= \frac{3}{4}\ln\lvert x+1\rvert + \frac{5}{4}\ln\lvert x-3\rvert - \frac{6}{x-3} +$	$\frac{2x^2-x+9}{(x+1)(x-3)^2} = \frac{A}{x+1} + \frac{B}{x-3} + \frac{C}{(x-3)^2}$ $2x^2-x+9 = A(x-3)^2 + B(x+1)(x-3) + C(x+$ Let $x=3$, $24 = 4C \Rightarrow C = 6$ Let $x=-1$, $12 = 16A \Rightarrow A = \frac{3}{4}$ Let $x=0$, $9 = \frac{3}{4}(9) - 3B + 6 \Rightarrow B = \frac{5}{4}$ $\therefore \frac{2x^2-x+9}{(x+1)(x-3)^2} = \frac{\frac{3}{4}}{x+1} + \frac{\frac{5}{4}}{x-3} + \frac{6}{(x-3)^2}$

5. (a) $\int \frac{1}{\sqrt{4-(x-3)^2}}\ \mathrm{d}x$ (b) $\int \frac{2}{x^2-6x+8}\ \mathrm{d}x$ (c) $\int \frac{x}{x^4-4}\ \mathrm{d}x$

(d) $\int \frac{1}{x^2+2x+3}\ \mathrm{d}x$ (e) $\int \frac{1}{\sqrt{6-4x-x^2}}\mathrm{d}x$

a) $\int \frac{1}{\sqrt{4-(x-3)^2}}\ \mathrm{d}x = \sin^{-1}\frac{x-3}{2} + c$
b) $\int \frac{2}{x^2-6x+8}\ \mathrm{d}x = \int \frac{2}{(x-3)^2-1}\ \mathrm{d}x = 2\left(\frac{1}{2}\right)\ln\left\lvert\frac{(x-3)-1}{(x-3)+1}\right\rvert + c = \ln\left\lvert\frac{x-4}{x-2}\right\rvert + c$
c) $\int \frac{x}{x^4-4}\ \mathrm{d}x = \frac{1}{2}\int \frac{2x}{\left(x^2\right)^2-4}\ \mathrm{d}x = \frac{1}{8}\ln\left\lvert\frac{x^2-2}{x^2+2}\right\rvert + c$
d) $\int \frac{1}{x^2+2x+3}\ \mathrm{d}x = \int \frac{1}{(x+1)^2+2}\ \mathrm{d}x = \frac{1}{\sqrt{2}}\tan^{-1}\frac{x+1}{\sqrt{2}} + c$
e) $\int \frac{1}{\sqrt{6-4x-x^2}}\mathrm{d}x = \int \frac{1}{\sqrt{-(x^2+4x-6)}}\mathrm{d}x = \int \frac{1}{\sqrt{-\left[(x+2)^2-10\right]}}\mathrm{d}x$ $= \int \frac{1}{\sqrt{10-(x+2)^2}}\mathrm{d}x = \sin^{-1}\frac{x+2}{\sqrt{10}} + c$